

Problem-Solution Fit

Date	15 July 2026
Team / Submission ID	
Project Name	Emotion Detection & Learning Support Engine
Submitted By	Venu Gopal (Sole Team Member / Team Lead)
Maximum Marks	5 Marks

Problem-Solution Fit Canvas

1. Customer Segment(s) [CS]	9. Problem Root / Cause [RC]	5. Available Solutions [AS]
Self-paced learners on online / e-learning platforms — school and college students studying Computer Science, Mathematics, Physics and similar technical subjects independently.	Digital learning platforms only track clicks, scores and time-on-page. There is no mechanism to read the learner's affective state from what they actually type when describing a problem.	Generic chatbots (no emotion awareness), static FAQ / tutoring pages, one-size-fits-all hint systems, human tutors (not always available on demand).
2. Problems / Pains + Frequency [PR]	7. Behavior + Intensity [BE]	6. Customer Limitations [CL]
- Confusion or frustration goes unnoticed — happens on nearly every study session (daily). - Learner disengages or gives up silently instead of asking for help.	Learners repeatedly re-read material or search externally rather than seeking help; intensity rises the longer they stay stuck, often ending in the session being abandoned.	Limited time, no access to a live tutor at the moment of confusion, and reluctance to ask for help in front of peers (budget: free / low-cost tools only, device: laptop/phone browser).
3. Triggers to Act [TR]	10. Your Solution [SL]	8. Channels of Behavior [CH]
Getting stuck on a concept, receiving a low quiz score, or feeling bored/under-challenged after finishing easy material.	Emotion Detection & Learning Support Engine: a Streamlit app that classifies the learner's emotion from free text using BiLSTM + BERT, then uses Gemini (with a rule-based fallback) to generate a personalised, empathetic response and next step.	Online: browser-based Streamlit web app.
4. Emotions Before / After [EM]		
- Before: anxious, confused, frustrated, or bored and unsupported. - After: reassured, understood, and given a concrete next step — more motivated to continue.		